



Ref: SMTTP/MECH MAINT-/2024-25/LMI_FIRE/552007

Date: 01/10/2024(18:45:48)

Subject: Approval for LMI on 'Routine inspection of fire protection devices oil piping system to reduce potential fire risk in Turbo Generators'

CC-OS has revised the OD with ref.no: COS-ISO-00-D/OPS/MECH/004 Rev. No.: 3, Sep. 2024. The OD supersedes and merges OD/OPS/MECH/004 Rev 02, Mar 2022 - "Routine Inspection of Turbine Control Oil Pipe work to reduce Potential Fire Risk" and OD/OPS/MECH/009 Rev 02, Aug 2022 - "Routine Inspection of Fire Protection Devices and System to reduce Potential Fire risk in Turbo Generator".

So, the LMI on 'Routine inspection of fire protection devices, oil piping & system to reduce potential fire risk in Turbo Generators' with Ref no: LMI/OD/OPS/MECH/004/ Rev.-06 is due for revision and necessary changes are done in the LMI.

LMI is hereby put up for approval of Competent Authority.

Changes incorporated in this revision (Rev: 6), have been mentioned as below

SI No	LMI/OD/OPS/MECH/004/ ISSUE-1, REV.-05, Mar 2022	LMI/OD/OPS/MECH/004/ ISSUE-1 REV.-06, Sept 2024
1.	LMI/OD/OPS/MECH/004/ ISSUE-1, REV.-05, Mar 2022 "Routine inspection of steam turbine oil pipework to reduce potential fire risk" LMI/OD/OPS/MECH/009/ ISSUE-1, REV.-06, Sept 2022 "Routine inspection of fire protection devices to reduce potential fire risk in BHEL sets"	TC document COS-ISO-00-OD/OPS/MECH/009 Rev 02 "Routine Inspection of Fire Protection Devices and System to reduce Potential Fire risk in Turbo Generator" TC document COS-ISO-00-OD/OPS/MECH/004 Rev 02 "Routine Inspection of Turbine Control Oil Pipe work to reduce Potential Fire Risk" Both OD merged as new revision of COS-ISO-00-OD/OPS/MECH/004 Rev. No. 3 "Routine Inspection of Fire Protection Devices, Oil Piping & System to reduce potential Fire Risk in Turbo Generator"

Submitted By :



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On:01/10/2024(19:05:44) Ref:SMTTP/MECH MAINT-/2024-25/LMI_FIRE/552007

Unauthorized Use Prohibited

Note No:2

LMI is reviewed in line with revised OD with ref.no: COS-ISO-00-OD/OPS/MECH/004 issued by CC-OS.
Put up for approval of competent authority.

Submitted by:

के पी रमेश - वरिष्ठ प्रबंधक

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Note No:3

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Note No:4

As discussed

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Note No:5

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Note No:6

May be approved

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Note No:7

Please review.

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Note No:8

reviewed

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Note No:9

LMI is in line with OD. May please approve.

Submitted by:

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Note No:10

May please be approved.

Submitted by:

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Note No:11

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On:23/10/2024(00:32:09) **Ref:**SMTTP/MECH MAINT-/2024-25/LMI_FIRE/552007

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

**Routine inspection of fire protection devices, oil
piping & system to reduce potential fire risk in
Turbo Generators**

LMI/OD/OPS/MECH/004/

SEPTEMBER-2024

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	Routine inspection of fire protection devices, oil piping & system to reduce potential fire risk in Turbo Generators
2	LMI No.	LMI/OD/OPS/MECH/004/ REV.-06 merges LMI/OD/OPS/MECH/004/ ISSUE-1, REV.-05, Mar 2022 LMI/OD/OPS/MECH/009/ ISSUE-1, REV.-06, Sept 2022
3	Reference Document	COS-ISO-00-D/OPS/MECH/004 Rev. No.: 3, Sep. 2024 Supersedes and merges. 1. OD/OPS/MECH/009 Rev 02, Aug 2022 2. OD/OPS/MECH/004 Rev 02, Mar 2022
4	Superseded Document	LMI/OD/OPS/MECH/004/ ISSUE-1, REV.-05, Mar 2022 LMI/OD/OPS/MECH/009/ ISSUE-1, REV.-06, Sept 2022
5	Prepared By	Name: Sahil Sankalp, Manager (TMD)
6	Checked By	Name: Nilanjan Mandal, AGM (TMD)
7	Reviewed By	Name: Prasanta Kumar Jena GM (O&M)
8	Approved By	Name: S. K. Sinha H O P (Simhadri)

INDEX

Sl. No.	Particulars	Page No.
1.0	Introduction	1
2.0	Superseded Documents	1
3.0	Details of Revision	1
4.0	Process	2
5.0	Records to be maintained	5
6.0	Review	6
7.0	Distribution of LMI	6
8.0	LMI Checklist	6
9.0	List of Annexures	8

TITLE	Routine inspection of fire protection devices, oil piping & system to reduce potential fire risk in Turbo Generators
LMI No.	LMI/OD/OPS/MECH/004/

1.0 INTRODUCTION

Instances of failure of oil system pipework of steam turbines due to vibration, rubbing and failure of welded or compensation joints are not uncommon and in certain instances have led to oil catching fire. Oil piping's are laid in close proximity to steam pipework and other hot surfaces, any leakage therefore poses a potentially severe fire risk & may cause outage of unit. On many occasions, leakages in these oil lines resulted in major fires causing severe damage to both turbines and station structures. The main objective of this document is to cover routine inspection of oil piping, fire protection devices to protect TG equipment from fire hazard.

2.0 SUPERSEDED DOCUMENTS

LMI/OD/OPS/MECH/004/ ISSUE-1, REV.-05, Mar 2022
LMI/OD/OPS/MECH/009/ ISSUE-1, REV.-06, Sept 2022

3.0 DETAILS OF REVISION

Following changes have been incorporated in the current revision –

SI No	LMI/OD/OPS/MECH/004/ ISSUE-1, REV.-05, Mar 2022	LMI/OD/OPS/MECH/004/ ISSUE-1, REV.-06, Sept 2024
1.	LMI/OD/OPS/MECH/004/ ISSUE-1, REV.-05, Mar 2022 "Routine inspection of steam turbine oil pipework to reduce potential fire risk" LMI/OD/OPS/MECH/009/ ISSUE-1, REV.-06, Sept 2022 "Routine inspection of fire protection devices to reduce potential fire risk in BHEL sets"	TC document COS-ISO-00-OD/OPS/MECH/009 Rev 02 "Routine Inspection of Fire Protection Devices and System to reduce Potential Fire risk in Turbo Generator" TC document COS-ISO-00-OD/OPS/MECH/004 Rev 02 "Routine Inspection of Turbine Control Oil Pipe work to reduce Potential Fire Risk" Both OD merged as new revision of COS-ISO-00-OD/OPS/MECH/004 Rev. No. 3 "Routine Inspection of Fire Protection Devices, Oil Piping & System to reduce potential Fire Risk in Turbo Generator"

TITLE	Routine inspection of fire protection devices, oil piping & system to reduce potential fire risk in Turbo Generators
LMI No.	LMI/OD/OPS/MECH/004/

4.0 PROCESS:

4.1 If not already built, the following rooms with separation walls and fire check doors need to be constructed (if envisaged in design) to contain oil spillage and fires.

a. Room around following equipment:

Main Oil Tank

Lube Oil Coolers

Oil purifier, Oil Filters

b. Valve room around Main steam, reheat stop and control valves.

4.2 Fire extinguishers of sufficient designated quantity for 'B' class fire (CO₂, Foam, DCP) are to be provided in the above areas as per OSHA Standard 1910.157 and IS: 15683 latest edition.

4.3 Only Flame proof light fitting should be provided in the MOT Room (as per IS: 2206).

4.4 Automatic actuation of fire protection in case of Main oil tank level low, (normally set at -100 mm in BHEL supplied KWU machines) should not be kept out of service while turbine oil system is in operation.

Joint protocol for MOT Zero position checking & calibration for all units shall be made by Turbine maintenance along with operation.

4.5 With regard to main oil tank level and turbine starting, no interlock, protection and annunciation to be bypassed and fire protection channel to be always kept in service even during barring gear operation.

4.6 Local fire protection push buttons should be labeled and covered such that the areas protected by push buttons and resulting actions are known to the Operating staff before actuation.

4.7 Periodic checking of functioning of Fire Protection devices by actuating local push buttons & from Unit Control Board and automatic fire protection system etc. to be done and recorded as per OD/OPS/SYST/007-Unit protection devices to ensure healthiness and actuation of various devices. Such checking should take place at not more than 1-year interval (as per IS: 2190) and before start of machine after every unit overhaul or any other available opportunity.

4.8 MOT fire protection sprinkler operation (as per OSHA 1910.163) checking shall be carried out as per CC-IOM COS/OPN/2020-21/O&M/, DATE: 19/01/2021. Main Oil Tank oil level normal in ATRS rolling to be confirmed. Oil level in MOT is to be maintained near normal level and overfilling not to be resorted to in order to avoid higher inventory of oil in MOT.

TITLE	Routine inspection of fire protection devices, oil piping & system to reduce potential fire risk in Turbo Generators
LMI No.	LMI/OD/OPS/MECH/004/

4.9 Periodic inspection of MOT room oil centrifuge antiflood tank area for unwanted oil spillage & oil accumulation to be done. Necessary cleaning & housekeeping had to be done, if required. Anti-flood tank level switch healthiness to be ensured in case of malfunction.

4.10 Main steam and Reheat stop and control valves and associated piping in TG hall should be inspected quarterly for vibrations. Checks for following is recommended:

- a. Visual inspection of supporting structure of HPT/IPT Stop and control valves
- b. Visual inspection for any leakages & missing fasteners.
- c. Integrity of hangers & supports system. Healthiness of vibration isolation system of supports. In case Engg Schematic is not available, a layout sketch may be prepared by station, indicating location of hangers and supports.
- d. Inspection of fouling of pipelines with hot surface & rubbing with another metallic surface.
- e. Proper illumination in the above area.
- f. Proper approach to above stop & control valves including servomotors.
- g. Any defect observed during visual inspection should be attended immediately or during next available opportunity depending upon the severity of the defect.

The inspection shall be carried out quarterly. Joint protocol shall be made by Turbine maintenance along with operation.

4.10 Critical piping hangers, supports, restraints and expansions are to be checked during every unit overhaul. Hanger / support cold & hot readings are to be recorded and any deviation observed shall be adjusted / rectified during every unit overhaul (Refer TC Doc No. OD-OPD-MECH-002: High Pressure Pipe Support System)

A Joint protocol shall be made by Turbine maintenance along with FQA.

After incidence of safety valves operations, any steam/water hammering, abnormal valve closure especially in any of the Main Steam Hot Reheat, and Cold Reheat lines, the restraints/hangers should be inspected along with expansions on turbine casings. In case of any breakage in the support system, the defect must be attended without delay.

Joint protocol shall be made by Turbine maintenance along with operation.

TITLE	Routine inspection of fire protection devices, oil piping & system to reduce potential fire risk in Turbo Generators
LMI No.	LMI/OD/OPS/MECH/004/

4.11. Details of routine inspection of oil piping

Routine visual inspection for leaks, vibration or inadequate support of all lube oil, jacking & control oil pipework should be carried out during a turbine run-up and at steady load operation by conducting walk down checks. Checks for following integral components is recommended:

- a. Visual inspection for any oil leakage, gland leakage & missing fasteners of flange connections.
- b. Integrity of hangers & supports system. Healthiness of vibration isolation system of supports.
- c. Inspection of fouling of pipelines with hot surface & rubbing with another metallic surface.
- d. Proper illumination in oil rooms & oil canal.
- e. Proper approach to oil canal with staircase.
- f. Any defect observed during visual inspection should be attended immediately or during next available opportunity depending upon the severity of the defect.

The inspection shall be carried out quarterly. Joint protocol shall be made by Turbine maintenance along with operation.

4.12. Details of inspection of oil piping during unit overhaul

During turbine overhauls, all lube oil, jacking oil & control oil pipework joints, whether welded or of other construction, should be visually inspected after proper cleaning. All pipes in the region of supports should be inspected for fretting or other damage. NDT should be made of any areas which appear suspect and as mentioned below.

- a. 100% DPT of weld joints of jacking oil system.
- b. DPT of 1/3rd weld joints for lube oil & control oil supply piping in phased manner so that all joints may be covered in three cycles.
- c. Any defect observed during DPT test should be rectified immediately.
- d. Any loose or missing support, hangers, or clamps to be rectified and suitable rubber pads to be provided in required places to prevent excessive vibration in pipeline or direct rubbing of pipeline in case of fouling.
- e. In case of close proximity to steam lines or hot surfaces, adequate measures to be

TITLE	Routine inspection of fire protection devices, oil piping & system to reduce potential fire risk in Turbo Generators
LMI No.	LMI/OD/OPS/MECH/004/

taken to prevent heating of oil pipelines either by pipe re-route or providing insulation lags.

f. A Joint protocol shall be made by Turbine maintenance along with FQA.

5.0 RECORDS TO BE MAINTAINED

Sl No.	Record Description	Maintained By	Frequency	Remarks
1	Records must be kept for all inspections and should contain the following data:	Responsibility of carrying out above inspection shall be with:	As mentioned Above.	Rectification of observed defects during available opportunity/ Unit overhaul
	Date of visual inspection of hangers & supports & supporting structure	TMD	Quarterly	
	Details of MOT fire protection sprinkler operation checking	Operation	During Unit Overhaul	
	Details of incidents of safety valves operations, any steam/water hammering, abnormal valve closure of turbine valves.	Operation	During occurrence	
	Exception of the inspection.	TMD		
	Details of rectification carried out.	TMD However, joint protocols shall be made along with TMD, Operation and FQA Dept as applicable.		

TITLE	Routine inspection of fire protection devices, oil piping & system to reduce potential fire risk in Turbo Generators
LMI No.	LMI/OD/OPS/MECH/004/

2	Records must be kept for all piping inspections and should contain the following data: Date of visual inspection Details of DPT Exception of the inspection	Responsibility of carrying out above inspection shall be with TMD Department. However, joint protocols shall be made along with Operation and FQA Dept as applicable.	As mentioned, Above.	Rectification of observed defects during available opportunity/ unit overhaul
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6.0 REVIEW

This LMI document shall be reviewed by the HOD-TMD on a 2 yearly basis or as necessary and the same shall be approved by HOP.

7.0 DISTRIBUTION OF LMI

Soft copy of the LMI shall be uploaded in the EEMG dept. website on Simhadri intranet LAN by the EEMG Department.

8.0 LMI CHECKLIST

Checklist for Routing inspection of fire protection devices, oil piping & system to reduce potential fire risk in Turbo generator

	Operation	Responsibility	Frequency	Remarks
1.	Ensure oil rooms & valve room are having separation walls and fire check doors with closing arrangement	Operation	Regular	
2.	Ensure availability of Fire extinguishers of sufficient designated quantity for 'B' class fire	Operation/Offsite maint.	Quarterly	
3.	Ensure Flame proof light fitting available in MOT Room	EMD	Quarterly	
4.	MOT Zero position checking &	Operation &	Overhaul:	

TITLE	Routine inspection of fire protection devices, oil piping & system to reduce potential fire risk in Turbo Generators
LMI No.	LMI/OD/OPS/MECH/004/

	calibration for all units	TMD	Every 02 years	
5.	No interlock, protection and annunciation to be bypassed and fire protection channel to be always kept in service even during barring gear operation	Operation	Regular	
6.	Local fire protection push buttons are labeled and Covered	Operation	Quarterly	
7.	MOT fire protection sprinkler operation checking is carried out as per CC-IOM COS/OPN/2020-21/O&M/, DATE: 19/01/2021	Operation	Overhaul: Every 02 years	
8.	Main Oil Tank oil level normal in ATRS rolling is being confirmed. Oil level in MOT is being maintained near normal level.	Operation	Regular	
9.	Hangers of Main steam and Reheat stop and control valves and associated piping in TG hall to be inspected	TMD	Quarterly	
10.	Routine visual inspection for leaks, vibration or inadequate support of all lube oil, jacking & control oil pipework is being carried out Proper illumination in oil rooms & oil canal.	TMD / Operation	Quarterly	
11.	Proper illumination in oil rooms & oil canal available	Operation/ EMD	Regular	
12.	Proper approach to oil canal with staircase available	TMD	Regular	
13.	During turbine overhauls, Ensure NDT of all lube oil, jacking oil & control oil pipework joints 100% DPT of weld joints of jacking oil system.	TMD & FQA	Overhaul: Every 02 years	

TITLE	Routine inspection of fire protection devices, oil piping & system to reduce potential fire risk in Turbo Generators
LMI No.	LMI/OD/OPS/MECH/004/

	DPT of 1/3rd weld joints for lube oil & control oil supply piping in phased manner so that all joints may be covered in three cycles.			
14.	Any loose or missing support, hangers, or clamps to be rectified and suitable rubber pads to be provided	TMD	Quarterly	
15.	In case of close proximity to steam lines or hot surfaces, adequate measures are taken to prevent heating of oil pipelines either by pipe re-route or providing insulation lag	TMD	Quarterly	
16.	After incidence of safety valves operations, any steam/water hammering, abnormal valve closure especially in any of the Main Steam Hot Reheat, and Cold Reheat lines, the restraints/hangers should be inspected along with expansions on turbine casings	TMD/ Operation	During occurrence	

9.0 LIST OF ANNEXURES

Annexure-I: Reference Documents

Annexure-II: Copy of IOM Procedure to be adopted in case of MOT (TG, TDBFP's) fire protection sprinkler operation checking.

Annexure-III: Walk down checklist

Annexure-IV: Protocol for DPT checking of control fluid line weld joints

Annexure-V: Protocol for DPT checking of lube oil line weld joints

Annexure-VI: Protocol for DPT checking of jacking oil line weld joints

TITLE	Routine inspection of fire protection devices, oil piping & system to reduce potential fire risk in Turbo Generators
LMI No.	LMI/OD/OPS/MECH/004/

ANNEXURE I**Reference Documents**

S. No	Description	Reference Document	Remarks
1.	Deployment of 'B' Type Fire Extinguisher	OSHA Standard 1910.157 IS: 15683 latest edition	
2.	Specification for flameproof electric lighting fittings	IS: 2206 latest edition	
3.	Fire Protection Fixed Sprinkler System	OSHA Standard 1910.163	
4.	Integrity of Hangers and Supports system for Lub Oil Piping	Stage I Drg. No. 0-13111-81001 Drg. No. 0-13111-81002 Drg. No. 0-13111-81003 Drg. No. 0-13111-81004 Drg. No. 0-13111-81005 Drg. No. 0-13111-81006 Stage II Drg. No. 0-13111-N3001 Drg. No. 0-13111-N3002 Drg. No. 0-13111-N3003 Drg. No. 0-13111-N3004 Drg. No. 0-13111-N3005 Drg. No. 0-13111-N3006	
5.	Periodic checking of all fire protection devices to be done in not more than 1-year interval	IS: 11062	
6.	Placement requirement for Class B Fire Extinguisher	IS: 11062	

TITLE	Routine inspection of fire protection devices, oil piping & system to reduce potential fire risk in Turbo Generators
LMI No.	LMI/OD/OPS/MECH/004/

ANNEXURE II



INTER OFFICE MEMO

FROM: GM (CC - OS – Operation)

TO:

GM (O&M)

All Coal Stations

REF: COS/OPN/2020-21/O&M/

CC:

HOP

All Coal Stations

DATE: 19/01/2021

Sub: Procedure to be adopted in case of MOT (TG, TDBFP's) fire protection sprinkler operation checking – Reg.

It is to inform that in one of our NTPC Stations, TG MOT Fire protection sprinkler, maloperated while unit is on bar and MOT level went up due to water entry into the MOT tank. In this regard, the following recommendations need to be implemented across all stations.

1. Procedure for MOT Fire protection sprinkler checking
 - a. MOT Fire protection sprinkler actual operation is to be checked during unit overhaul when MOT is empty.
 - b. This actual operation is to be checked without covering auxiliaries and tank with plastic or tarpaulin cloth. Necessary clearance from concerned maintenance agencies should be taken before operation of Fire protection sprinkler.
 - c. During Fire protection sprinkler operation, sources of water entry into the system to be identified and attended. If any water entry is there, Fire protection sprinkler operation to be repeated till the leakage points are fully attended.
 - d. A protocol to this is to be prepared with signatures from operation, turbine, C&I, EMD and safety.
2. Immediately after oil flushing in the overhaul, MOT top covers are to provided with necessary gaskets as per the system requirement and operation should ensure this before going ahead with further activities.
3. Any C&I instruments provided on top of MOT tank should be properly sealed and ensure that no water ingress is there from these points.
4. MOT Fire protection sprinkler operated alarm should be hooked up to Unit level alarm system in DCS, in addition to the alarm available at repeater panel in the standalone fire protection and alarm system. C&I to do necessary modification as required at the station.

All stations are requested to comply the same and send the compliance to the undersigned.

(V. KUMARASWAMY)

TITLE	Routine inspection of fire protection devices, oil piping & system to reduce potential fire risk in Turbo Generators
LMI No.	LMI/OD/OPS/MECH/004/

ANNEXURE-III: Walk down checklist

Sl No	Description of Particulars	Status	Exceptions, if any
1	Visual inspection for any oil leakage, gland leakage & missing fasteners of flange connections		
2	Integrity of hangers & supports system		
3	Healthiness of vibration isolation system of supports		
4	Inspection of fouling of pipelines with hot surface & rubbing with other metallic surface		
5	Proper illumination in oil rooms & oil canal		
6	Proper approach to oil canal with staircase		

Any Other Observations:

Signatures:

TMD:

Operation Dept:

TITLE	Routine inspection of fire protection devices, oil piping & system to reduce potential fire risk in Turbo Generators
LMI No.	LMI/OD/OPS/MECH/004/

ANNEXURE-IV: Protocol for DPT checking of control fluid line weld joints

S. No.	Description of pipeline/no. of joints	Date of testing

CHECKED & CERTIFIED BY:		
SIGNATURE		
NAME		
DESIGNATION		
DEPT	NTPC-TMD	NTPC-FQA

TITLE	Routine inspection of fire protection devices, oil piping & system to reduce potential fire risk in Turbo Generators
LMI No.	LMI/OD/OPS/MECH/004/

Annexure-V: Protocol for DPT checking of lube oil line weld joints

S. No.	Description of pipeline/no. of joints	Date of testing

CHECKED & CERTIFIED BY:		
SIGNATURE		
NAME		
DESIGNATION		
DEPT	NTPC-TMD	NTPC-FQA

TITLE	Routine inspection of fire protection devices, oil piping & system to reduce potential fire risk in Turbo Generators
LMI No.	LMI/OD/OPS/MECH/004/

Annexure-VI: Protocol for DPT checking of jacking oil line weld joints

S. No.	Description of pipeline/no. of joints	Date of testing

CHECKED & CERTIFIED BY:		
SIGNATURE		
NAME		
DESIGNATION		
DEPT	NTPC-TMD	NTPC-FQA